

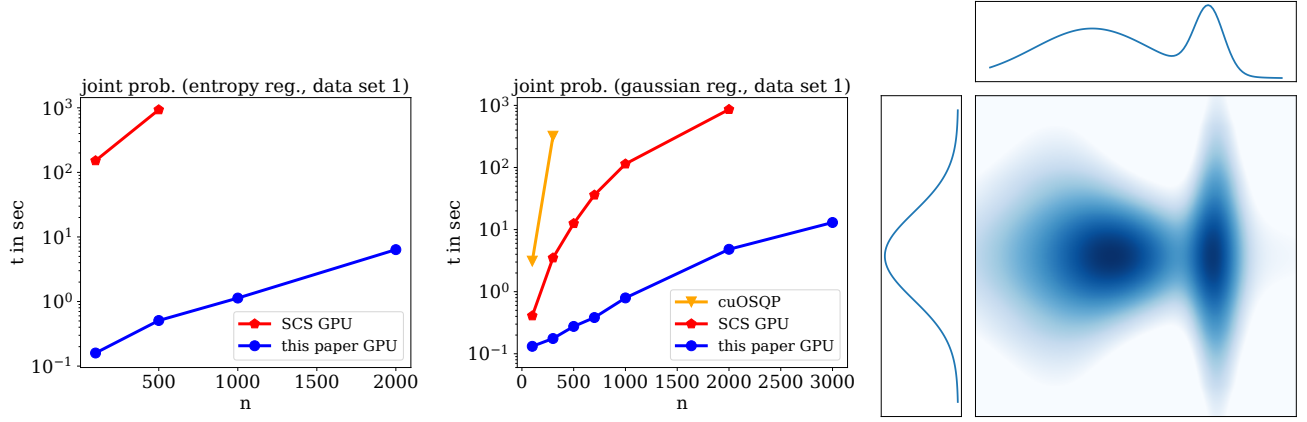


Figure 3: Running times for computing the joint probability distribution from two marginal distributions. The left plot shows the running times when the entropy prior is used and the plot in the middle when a Gaussian prior is used. The right figure visualizes the probabilities.

and cuOSQP into the experiments once their running time exceeded 1000 seconds. It can be seen that SCS is faster than cuOSQP on the first set of experiments and slower than cuOSQP on the second set. However, our approach outperforms SCS and cuOSQP by several orders of magnitude in both sets of experiments.

#### 5.4 Joint Probability Distribution

Given two discrete probability distributions  $u \in \mathbb{R}^m$  and  $v \in \mathbb{R}^n$ , we are interested in their joint probability distribution  $P \in \mathbb{R}^{m \times n}$ . This problem has been studied intensively before, see, e.g., (???). With additional knowledge, it can be reduced to a regularized optimal transport problem. Many different regularizers have been used, for instance, an entropy, a Gaussian, or more generally, a Tsallis regularizer. The corresponding optimization problem is the following constrained optimization problem over positive matrices

$$\min_P \langle M, P \rangle + \lambda r(P) \quad \text{s. t.} \quad P \cdot \mathbf{1} = u, P^\top \cdot \mathbf{1} = v, 0 \leq P,$$

where  $M \in \mathbb{R}^{m \times n}$  is the cost matrix,  $r(\cdot)$  is the regularizer,  $\mathbf{1}$  is the all-ones vector, and  $\lambda \in \mathbb{R}_+$  is the regularization parameter. In our experiments we used the entropy and the Gaussian regularizer that are both special cases of the Tsallis regularizer. In the special case that the regularizer is the entropy,  $m = n$ , and the cost matrix  $M$  is a metric, (?) showed that the problem can be solved using Sinkhorn's algorithm (?). Similar results are known for other special cases (?). However, in general Sinkhorn's algorithm cannot be used as it is the case in the present experiments, since the cost matrix is not a metric.

Here, we used synthetic data sets since the real-world data sets that are usually used for this task are very small, typically  $m, n \leq 20$ . We created two sets of synthetic data sets. For the first set of data sets, we let a Gaussian and a mixture of two Gaussians be the marginals, see Figure 3. Then, we discretized both distributions to obtain the marginal vectors  $u$  and  $v$ . In this case, we set  $m = n$ . Hence, when  $n = 1000$ ,

the corresponding optimization problem has  $10^6$  optimization variables with lower bound constraints and 2000 equality constraints. The cost matrix  $M$  was fixed to be the discretized version  $uu^\top$  of a two-dimensional Gaussian, and the regularization parameter was set as  $\lambda = \frac{1}{2}$ . We ran two sets of experiments on this data set, one where  $r(\cdot)$  is the entropy regularizer and another one with the Gaussian regularizer. Figure 3 shows the running times for both experiments for varying problem sizes. It can be seen that our approach outperforms cuOSQP and SCS by several orders of magnitude. The cuOSQP solver ran out of memory already on very small problems. The second data set was created as in (?). On this data set, the speedup over cuOSQP and SCS is even more pronounced. Detailed results can be found in the appendix.

## 6 Conclusion

We presented an approach for solving constrained optimization problems on the GPU efficiently and included this approach into the GENO framework. The framework allows to specify a constrained optimization problem in an easy-to-read modeling language and then generates solvers in portable Python code that outperform competing state-of-the-art approaches on the GPU by several orders of magnitude. Using GPUs also for classical, that is, non-deep, machine learning becomes increasingly important as hardware vendors started to combine CPUs and GPUs on a single chip like Apple's M1 chip.

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